Band-1 Hadamard Order 8452 — Miyamoto (Paley-I Seed H_2112 q=2111 + C_2114); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_2112 Paley-I PASS; C_2114 PASS; H_8452 Miyamoto PASS; classical SPK FAIL → Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8356_PaleyII`; `2026-09-16_Band1_Hadamard_8116_PaleyII`; `2026-09-16_Band1_Hadamard_7988_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8452_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2112 and 8452 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{2112} = Paley-I($q = 2111$) and symmetric Paley conference C_{2114} over GF(2113) ($q = 2113 \equiv 1 \pmod 4$). Classical SPK for 8452: Paley-I **FAIL**s ($q = 8451 = 3^3 \cdot 313$ composite); Paley-II **FAIL**s ($q = 4225 = 5^2 \cdot 13^2$ NOT prime power — no matrix attempt); Kronecker **FAIL**s (4×2113). Therefore order **8452 is a Miyamoto-only bonus NOT in SPK**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8356 IN SPK via Paley-II** (seeds **2088+8356** folded). Prior SPK seeds **2088+8356 / 2028+8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2112** folded into ONBOX/SPK when verified; **8452** never folded (Miyamoto alone never folds SPK).

Abstract

Finite constructions of orders **2112** and **8452**.

- H_{2112} = Paley-I ($q = 2111$) — preferred seed (Kronecker 4×528 alternate eligible unused; Paley-II $q = 1055$ FAIL; Double FAIL)
- C_{2114} = Paley conference over GF(2113), $q = 2113$
- H_{8452} = Miyamoto(H_{2112}, C_{2114})
- Paley-I for 8452: **SKIP** ($q = 8451$ composite)
- Paley-II for 8452: **SKIP** ($q = 4225$ NOT prime power; no matrix attempt)
- Kronecker for 8452: **SKIP** ($4 \cdot 2113$ not admissible)

SPK: 8452 Miyamoto-only NOT in SPK. Seed 2112 enters SPK (Paley-I). Prior SPK **2088+8356 / 2028+8116 / 1996 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8452.py`

Results. overall_pass=true; units 146/146; wall 491.2617s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8452 = 4 \cdot 2113$; 2113 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{2112} and C_{2114} (Paley conference over prime field $q = 2113$).
- Seed: Paley-I $q = 2111$ (prime $\equiv 3 \pmod 4$) → H_{2112} (Kronecker 4×528 alternate eligible; Paley-II $q = 1055 = 5 \cdot 211$ FAIL; Double FAIL).
- SPK: Paley-I $q = 8451$ **FAIL**; Paley-II $q = 4225$ **FAIL** (NOT prime power); Kronecker **FAIL** → **Miyamoto-only NOT in SPK**.
- Prior: 8356 IN SPK via Paley-II shipped; this ship $q = 2113 \rightarrow n = 8452$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_2112 Paley-I	Paley-I q=2111 DIRECT	0	6.5481
C_2114	Paley conf GF(2113) q=2113	0	5.8205
H_8452 Miyamoto	Miyamoto(H_2112 Paley-I, C_2114)	0	473.419
H_8452 Paley-I	SKIP q=8451=$3^3 \cdot 313$	—	—
H_8452 Paley-II	SKIP q=4225=$5^2 \cdot 13^2$ (no matrix)	—	—
H_8452 Kronecker	SKIP $4 \cdot 2113$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **8356** lineage (**2088+8356**; **2028+8116**; **1996**; **1932**; **7604**; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 8452/7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **2112** (Paley-I); **exclude** Miyamoto-only bonuses including **8452**. Known count 1668. Next constructive gap **2052** (OPEN).

Quantity	Value
8452 in SPK	**False** (Miyamoto-only)
2112 in SPK	**True** (Paley-I q=2111)
2088 / 8356 / 2028 / 8116 in SPK	**True** (prior)
8452 / 7988 / 7732 / 7492 in SPK	**False** (Miyamoto-only; excluded)

Honesty gates

- `clay_closed = false`
- GATE-04 collection **LOCKED**; no invented J
- Finite-order constructions **2112** and **8452** ONLY (verified max|HH −nI|=0)
- **NO** full Hadamard conjecture claim; **NO** finished-ToE; **NO** RH / P NP / YM / Frankl / NS-solved / AC-solved
- SPK membership only via classical Paley-I / Paley-II / Kronecker (**Miyamoto alone is NOT SPK**)
- Standard constructive gap **2052** remains **OPEN**

Next optional

- **q=2129→n=8516** Double(PaleyI half=1064),Kronecker(4×532) OR **q=2141→n=8564** PaleyII(q=1069) OR method-attack SPK gap **2052** OR Track B / Object 1 meter await